

2(a)(i)	any two from: I repair cells 1 to, replace (old / worn-out / dead / short-lived) cells / repair (damaged) tissue ; 2 stem cells divide and, differentiate / become specialised ; A form different types of cells 3 <i>idea of self-renewal</i> e.g. stem cells divide to maintain the, pool / number, of stem cells ; I increase in number of stem cells 4 AVP ; <i>ref. to</i> movement of food through intestine / presence of pathogens, increases chance of damage cells (produced by stem cells) are genetically identical so the function of, the intestine / organs / tissues, can continue / AW <i>idea that</i> allows growth (of small intestine) during development / AW R growth of cells	2
2(a)(ii)	anaphase ;	1
2(a)(iii)	cytokinesis ;	1
2(b)	any two from 1 (two) sister / identical, <u>chromatids</u> attached together ; R 'sister chromosomes' A holds the chromatids together after, replication / S phase 2 attach (chromatids / chromosomes) to spindle (fibres) ; <i>in context of mitosis</i> R if stated in interphase A kinetochore <i>or</i> spindle microtubules for spindle (fibres) 3 (centromeres) divide / separate, (at the start) in anaphase ; A end of metaphase 4 so daughter chromosomes can be moved to opposite poles ; A sister / identical, chromatids <i>do not award MP4 unless MP3 already given</i>	2

Question	Answer		Marks
2(c)	<i>look for function in explanation if not on first line</i> <i>for each cell – one mark for the function and one mark for the explanation</i> <i>mark each cell straight through to max 2</i>		6
	cell	function	explanation
	P	absorption / uptake / movement in, of, (named) substances / nutrients	microvilli, to give a large surface area / increase surface area / increase number of transport proteins (for uptake of digested food)
		diffusion of, (named) substances / nutrients	I cilia have large surface area A large surface area:volume ratio for microvilli
		active uptake / active transport, of (named) substances / of nutrients	mitochondria to provide, ATP / energy, for, active transport / active uptake R produce / make, energy R energy for cilia
		breakdown of food molecules	large surface of membrane for enzymes for digestion
		production / secretion, of enzymes	microvilli for membrane enzymes / RER for making enzymes / Golgi for packaging enzymes or modification of enzymes
	Q	secretion / release / exocytosis / AW, of mucin / mucus I produce and provide	rough endoplasmic reticulum for making protein (part of mucin / mucus)
			Golgi body for, packaging / glycosylating / modification / AW, mucin A mucus AW = adding carbohydrate / sugars
		<i>if Q is secreting protein / unnamed substance allow ECF for explanation</i>	(large) vesicles / vacuoles, for storage / for containing / for transport, of mucin A mucus
	R	secretion / release / exocytosis / move ment out / AW, of (named) protein(s) I produce and provide A hormones / peptides / enzymes	(large quantity of) RER for, making proteins / making enzymes / translation / AW
			Golgi body for packaging / modification / AW, proteins
			vacuoles / vesicles, for transport to cell surface membrane / release by exocytosis / fusion with cell surface membrane